

School Math Workbook Targeted Mastery Worksheet (Set 12)

Module: Number Theory · Prime Factorization · Fractions Operations & 3D Geometry

Name: _____

Class: _____

Date: _____

Total: 100
pts

Score: _____

Section 1: Factors, Multiples & Primes

2 pts each, 20 pts total

1. List all positive factors in ascending order without omissions:

(1) All positive factors of 28: _____ ;

(2) All positive factors of 48: _____ .

2. Among the numbers 77, 83, 91, 97, the prime numbers are _____ , and the composite numbers are _____ .

3. Understand the meaning of powers and evaluate:

$$2^3 = \underline{\quad\quad} \quad 2^4 = \underline{\quad\quad} \quad 3^4 = \underline{\quad\quad} \quad 10^0 = \underline{\quad\quad} \quad 10^4 \times 10^5 = 10^{\underline{\quad\quad}}$$

4. The area of a square with an integer side length is called a square number. Among the integers from 1 to 20, the square numbers are _____ ; the number 6 is _____ a square number (write "is" or "is not").

5. Determine whether the outcome of each expression is Odd or Even:

(1) 3^2 is an _____ number (2) 6^2 is an _____ number (3) If n is even, $n^2 + 1$ is an _____ number

6. Consider the 5-digit number $\overline{4271\square}$: if it is divisible by 2, the digit $\square$ can be _____ ; if it is divisible by 3, the digit $\square$ can be _____ ; if it is divisible by both 2 and 5, the digit $\square$ must be _____ .

7. Express the prime factorization of each number in Index Form:

$$81 = \underline{\quad\quad\quad} \quad 72 = \underline{\quad\quad\quad}$$

8. Given two positive integers $E = 2^{10} \times 3^4$ and $F = 2^6 \times 3 \times 5^5$. In index form, their Highest Common Factor (HCF) is _____ ; their Lowest Common Multiple (LCM) is _____ .

9. The HCF of two positive integers a and b is 6, and their LCM is 90. If $a = 18$, then $b = \underline{\quad\quad\quad}$.

10. For $52 \div 24$, the quotient written in division-with-remainder form is _____ , and written as a mixed number in its simplest form is _____ .

Section 2: Prime Factorization & HCF/LCM

20 pts total

1. Use short division (divisors must be prime numbers) to perform prime factorization, and write out the final prime product (4 pts each, 8 pts total):

(1) Prime factorization of 54

[54 = _____]

(2) Prime factorization of 1080

[1080 = _____]

2. Use short division to find the HCF and LCM for each set of numbers (4 pts each, 8 pts total):

(1) Find HCF and LCM of 54 and 90

[HCF = _____, LCM = _____]

(2) Find HCF and LCM of 24, 36 and 48

[HCF = _____, LCM = _____]

3. Counting Factors Investigation (4 pts):

The prime factorization of 72 is $72 = 2^3 \times 3^2$. Use the prime exponent law to calculate how many positive factors 72 has in total. Show the calculation expression.

[Expression: _____ , Total factors: _____]

Section 3: Fractions Operations

20 pts total

1. Mental calculations and conversions (1 pt each, 4 pts total):

(1) $\frac{23}{6} = \underline{\hspace{2cm}}$
(mixed)

(2) $2\frac{3}{7} = \underline{\hspace{2cm}}$
(improper)

(3) $\frac{3}{6} + \frac{5}{6} = \underline{\hspace{2cm}}$
(simplest mixed)

(4) $1 - \frac{5}{12} = \underline{\hspace{2cm}}$

2. Addition and subtraction of unlike fractions (show common denominators, 2 pts each, 8 pts total):

① $\frac{2}{3} + \frac{1}{5}$

[Ans: _____]

② $2\frac{1}{3} + 3\frac{5}{6}$

[Ans: _____]

③ $4\frac{1}{4} - 1\frac{2}{3}$

[Ans: _____]

④ $\frac{5}{8} - \frac{1}{6} + \frac{7}{12}$

[Ans: _____]

3. Fraction multiplication and division (2 pts each, 8 pts total):

① $\frac{5}{12} \div \frac{15}{16}$

[Ans: _____]

② $\frac{7}{18} \times \frac{9}{14} \div \frac{5}{6}$

[Ans: _____]

③ $\left(\frac{3}{4} + \frac{1}{6}\right) \times 12$

[Ans: _____]

④ $\frac{5}{9} \div \left(\frac{2}{3} - \frac{1}{4}\right)$

[Ans: _____]

Section 4: 3D Spatial Geometry & Blocks

20 pts total

1. Fundamental Rectangular Prism Calculations (6 pts):

A rectangular prism has length 5 cm, width 4 cm, and height 3 cm.

- (1) Calculate the total surface area of this rectangular prism in cm^2 .
- (2) Calculate the volume of this rectangular prism in cm^3 .

[(1) Surface Area: _____ cm^2 ; (2) Volume: _____ cm^3]

2. Stacking and Joining Identical Prisms (6 pts):

There are two identical rectangular prisms, each measuring 5 cm by 3 cm by 2 cm. They are joined along one entire face to form a single larger solid.

- (1) What is the total surface area of the combined solid if the glued face has maximum area?
- (2) What is the total surface area of the combined solid if the glued face has minimum area?

[(1) Max glued face area solid: _____ cm^2 ; (2) Min glued face area solid: _____ cm^2]

3. Tiered Composite Block Model (8 pts):

Unit cubes with edge length 1 cm are arranged on a tabletop to build a 3-tier stepped solid: the bottom tier is a 3×3 grid (9 cubes); the middle tier is a 2×2 grid placed flush against one corner (4 cubes); the top tier consists of 1 cube flush against that same corner.

- (1) How many unit cubes are used in total? What is the total volume of the solid in cm^3 ?
- (2) All exposed surfaces (excluding the bottom resting on the table) are painted. What is the total painted surface area in cm^2 ?

[(1) Total cubes: _____ , Volume: _____ cm^3 ; (2) Painted Area: _____ cm^2]

Section 5: High-Order Problem Solving

5 pts each, 20 pts total

1. Divisibility Rules & Multiple Branches (5 pts):

The 6-digit number $\overline{5a342b}$ is divisible by 2, 3, and 5 simultaneously.

- (1) What is the value of the last digit b ?
- (2) Find all possible values of digit a .

[(1) $b =$ _____ ; (2) All possible values of a : _____]

2. Statistical Measures: Mean, Median, Mode & Range (5 pts):

A basketball player scored the following points in 10 consecutive games:

24, 28, 27, 31, 27, 35, 27, 29, 31, 41

- (1) Find the Range and the Mode of this dataset.
- (2) Find the Median and the Mean of this dataset.

[(1) Range: _____ , Mode: _____ ; (2) Median: _____ , Mean: _____]

3. Periodic Cycle & Starting Point (5 pts):

A playlist of 5 songs plays continuously in order: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A \rightarrow \dots$

- (1) If the playback starts from song A , which song is played 43rd?
(2) If song C is currently playing (counted as the 1st song), which song is played when the count reaches 38?

[(1) The 43rd song is _____ ; (2) The 38th song is _____]

4. Encounter Journey & Fraction Modeling (5 pts):

Person A and Person B start simultaneously from two points 1000 m apart and travel towards each other. Person A walks at 60 m/min, and Person B rides a bicycle at 140 m/min.

- (1) After how many minutes will they meet?
(2) At the moment they meet, what fraction of the total distance has Person A traveled? How many more meters has Person B traveled than Person A?

[(1) Meet _____ min; (2) Person A _____ of total, Person B _____ m more after traveled traveled]